

	Case Name: Convertible Roofing Project Owner: Eder Omar Vilca Carrillo	Sector	Construction
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Processed by	Amin Azimi		One of nine std. scenarios
Date	2025	Project Control	User defined distributions
File Name	C2025-8		Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

Construction of permanent convertible roofs to enhance infrastructure flexibility in educational or public spaces. The project includes site preparation, preliminary works, and full construction of adaptable roofing systems.

The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	75	Serial/Parallel (SP)	43%
Planned Duration (PD)	60 days	Activity Distribution (AD)	41%
Budget At Completion (BAC)	349,558.26€	Length of Arcs (LA)	0%
Renewable Resources	6 Types	Topological Float (TF)	38%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11.33	9.06	0.70
CRI-rho	11.49	8.74	0.76
CRI-tau	9.00	5.44	0.70

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	33.55	3.30	5.14
CRI-rho	33.55	3.30	5.14
CRI-tau	33.55	3.30	5.14

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	42.83	47.41	0.30
SI	52.27	42.65	0.05
SSI	6.88	10.78	2.16
CRI-r	10.99	9.39	1.89
CRI-rho	10.91	9.44	1.80
CRI-tau	9.79	7.18	0.83

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	4.83	1.01
PV - SPI	7.75	2.72
PV - SCI	7.86	2.74
ED - 1	4.77	1.06
ED - SPI	7.58	2.54
ED - SCI	7.66	2.55
ES - 1	3.92	0.91
ES - SPI(t)	6.17	2.70
ES - SCI(t)	6.28	2.70

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.24	0.00
CPI	0.42	0.00
SPI	4.23	0.00
SPI(t)	3.96	1.60
SCI	4.44	1.75
SCI(t)	4.17	1.60
0.8 CPI + 0.2 SPI	1.18	1.74
0.8 CPI + 0.2 SPI(t)	1.11	0.51

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **two** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0.00	-43235.79	-72	1	0.79	0.76	0.95
std dev	0.00	61144.64	56.57	1	0.30	0.26	0.07
final	0.00	0.00	-32	1	1.00	0.94	1.00

2.3.3.2. Time forecasting

PD	60 days	Real Duration	65 days	Delay	8%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	68.00	11.31	12.31	-4.62
PV - SPI	82.00	31.11	33.85	-26.15
PV - SCI	82.00	31.11	33.85	-26.15
ED - 1	69.00	7.07	7.69	-6.15
ED - SPI	84.00	28.28	30.77	-29.23
ED - SCI	84.00	28.28	30.77	-29.23
ES - 1	69.00	7.07	7.69	-6.15
ES - SPI(t)	84.50	28.99	31.54	-30.00
ES - SCI(t)	84.50	28.99	31.54	-30.00

2.3.3.3. Cost forecasting

BAC	349,558.26€	Real Cost	349,558.26€	On Budget	0%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	349558.28	0.00	0.00	0.00
CPI	349558.28	0.00	0.00	0.00
SPI	432186.41	116853.82	23.64	-23.64
SPI(t)	434103.83	119565.46	24.19	-24.19
SCI	432186.41	116853.82	23.64	-23.64
SCI(t)	434103.83	119565.46	24.19	-24.19
0.8 CPI + 0.2 SPI	360043.39	14828.18	3.00	-3.00
0.8 CPI + 0.2 SPI(t)	360196.47	15044.67	3.04	-3.04